Second mandatory assignment

Deadline for submission is April 30th. Submit by email or through Canvas. Presentation will be given in Class on April 30th.

The problems below are all related to the educational Boussinesq model *sbouss:* https://geirkp.github.io/bouss/index.html, which require numerical applications. Throughout these problems we assume that the computational domain is $0 \leq x \leq L$, where the length of the numerical wave tank, L, is either given or chosen appropriately by the user. Other parameters are also either given or must be chosen by the student. The choice of parameters in numerical simulations is an important task and may often involve both trial and error and simplified analysis. The simulations start with either a solitary wave solution, for which the *sbouss* program provides initial values for both u and η, or an initial elevation, specified through an input file, combined with zero initial velocity.

Problem 1: Solitary wave propagation and grid effects

We will test the performance of the model by investigating how well the shape of a solitary wave is preserved. Which model (LSW, NLSW, linear dispersive or Boussinesq) is the appropriate choice for this task? Use the *sbouss* program to simulate the propagation of a solitary wave, with amplitude $\alpha = 0.1$, starting at $x = 20$ until $t = 70$ on constant depth $h = 1$. Make certain that the numerical wave tank is long enough. Use Matlab, or some other tool, to find the maximum η at $t = 70$ when you choose $\Delta x = 2$, $\Delta x = 1$ and $\Delta x = 0.5$, while the time step reduction factor is set to 0.5 and the discrete correction term is turned off. Do also depict all three solutions in the same diagram. Comment on what resolution that seems relevant.

Problem 2: Wave dispersion

Use Matlab, for instance, to produce an initial elevation:

$$\eta(x,0) = \begin{cases} 2A\cos^2\left(\frac{\pi(x-x_0)}{\lambda}\right) & -\lambda/2 \leq x - x_0 \leq \lambda/2 \\ 0 & otherwise \end{cases}$$

which is combined with $u(x,0) = 0$. Explain why this initial condition has continuous first derivative and why it to leading order yields a wave in each direction with amplitude A and length λ.
Choose $A = 0.2$, $\lambda = 20$, $x_0 = 0$, a tank length $L = 100$ and simulation time $t = 80$. Remember that the wave tank starts at $x = 0$. We still have constant depth $h = 1$.

a) Solve the LSW equations. What should the result be? Adjust the resolution to obtain this result well enough (explain what you mean).
b) Solve the linearized Boussinesq equations. Plot the solution at $t = 80$. Discuss the solution in terms of results given in the course.
c) Solve the full Boussinesq equations. What is the difference from the previous case? What may the leading wave be? Depict the solution together with the previous one.
d) Solve the NLSW equations. You will see some short features. Use grid refinement to see if they change dramatically. If they do they are artifacts and the solution is invalid.

e) Solve the NLSW equations again, but focus on the times before the artifacts appear. What happens to the shape? Illustrate with figures. What may be the physical interpretation of this shape change?

Problem 3: Bores

We start from rest and use the same initial elevation as in Problem 2 with $x_0 = 0$. However, this time we employ a long initial condition and a long wave tank according to $L = 1000$ and $\lambda = 100$. In the two first subproblems we have constant depth $h = 1$.

a) We focus on $t < 200$. Do simulations with NLSW and Boussinesq equations. How long are the results similar? Illustrate with graphs.

b) Do the Boussinesq simulation until $t = 800$. What happens? Use Matlab to compare the second crest from the front to the solitary wave solution in the compendium. Can you expect perfect agreement ? How can the *sbouss* program be used to make a more accurate comparison?

c) A sequence of crests like the ones in the previous sub-problem is called an *undular bore*. It may also be generated due to shoaling. Make a depth file to *sbouss* which corresponds to $h = 1$ for $x < 40$, $h = 0.2$ for $150 > x > 50$ and a linear slope in between. Run a solitary wave with amplitude $A = 0.05$ from the deep to the shallow region (you need a fine grid due to the shallow shelf). Show that an undular bore is generated. Explain the relation to the previous sub-problem.